

BIKE STATION SHARING ANALYSIS

1. PROJECT OVERVIEW:

The Bike Station Sharing Analysis Dashboard is designed to provide a comprehensive view of bike-sharing operations across multiple cities and contractors. It consolidates key performance indicators (KPIs) such as station availability, bike distribution, and contractor contributions into a single interactive platform.

2. OBJECTIVES:

These are the objective were going to found from this analysis

- Identify the busiest bike-sharing stations across Europe to optimize resource allocation and maintenance schedules.
- Monitor operational health by tracking the proportion of open versus closed stations, ensuring service reliability.
- Compare bike distribution across contractors and cities to evaluate regional performance and demand.
- Analyse growth trends in bike-sharing infrastructure from 2022–2025 to forecast future expansion needs.
- Visualize the geographic spread of bike stations across Europe to identify coverage gaps and expansion opportunities.

3. DATA SOURCES:

Time line:2022-2024.

Domain: Europe bike station sharing.

4. DATA CLEANING (PREPROCESSING):

The following preprocessing steps were performed to ensure data quality:

- Handled missing values
- Removed duplicate records
- Corrected and changed data types
- Renamed columns for better readability
- Delimit the columns

- Sort and filter the column
- Ensured uniqueness across key identifiers for reliable aggregation.

TOOLS USED:

- Excel
- Power BI (power query)

5. DATA TRANSFORMATION:

To enhance analysis, the following transformations were applied:

- Created a dedicated Calendar Table to enable robust time-based analysis
- Developed custom columns to enhance analytical depth and improve reporting accuracy.
- Derived key performance indicators (KPIs).
- Derived longitude and latitude by splitting the position column.

6. DATA MODELLING AND DAX (POWER BI):

Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.

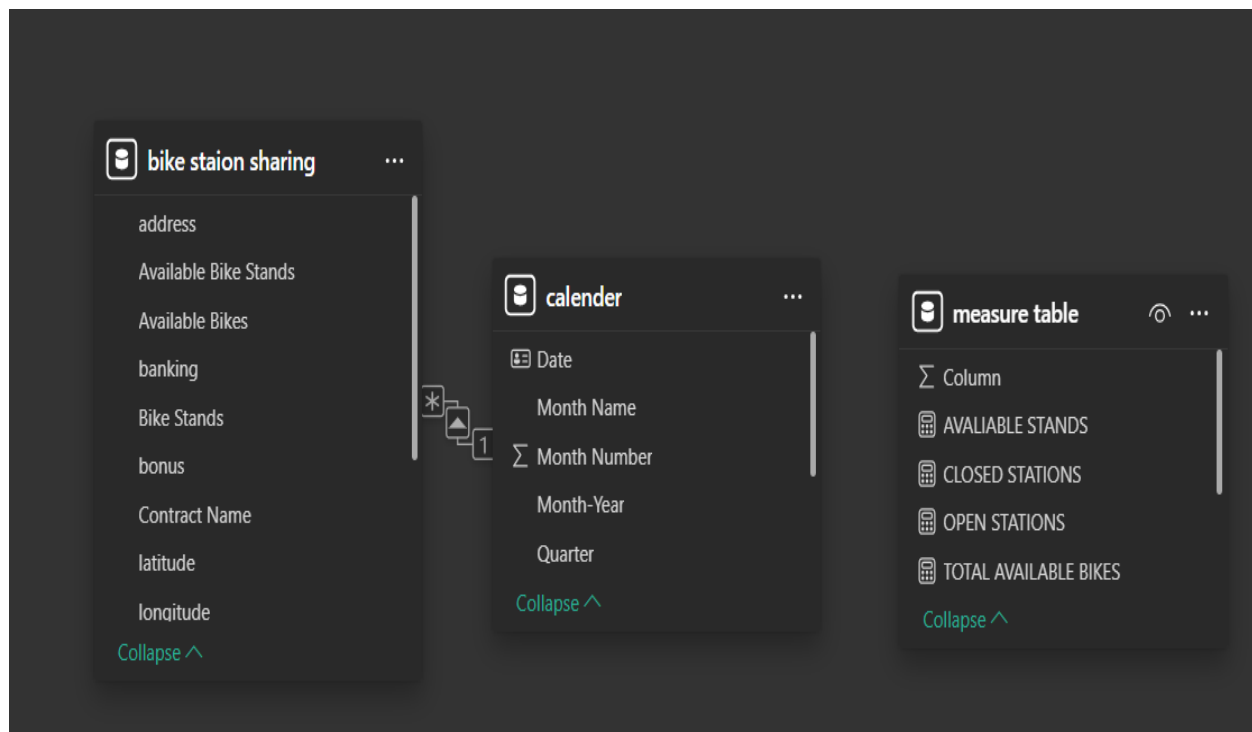


FIG:DATA MODELLING BETWEEN THE TABLES

DAX Measures:

These are the key measures identified in this analysis

- **Total Stations** → 2.856K
- **Closed Stations** → 206
- **Open Stations** → 2.65K
- **Available Stands** → 32K
- **Total Bikes** → 57K
- **Available Bikes** → 20K

7. EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION:

- Performed statistical analysis
- Built interactive dashboards and visualizations

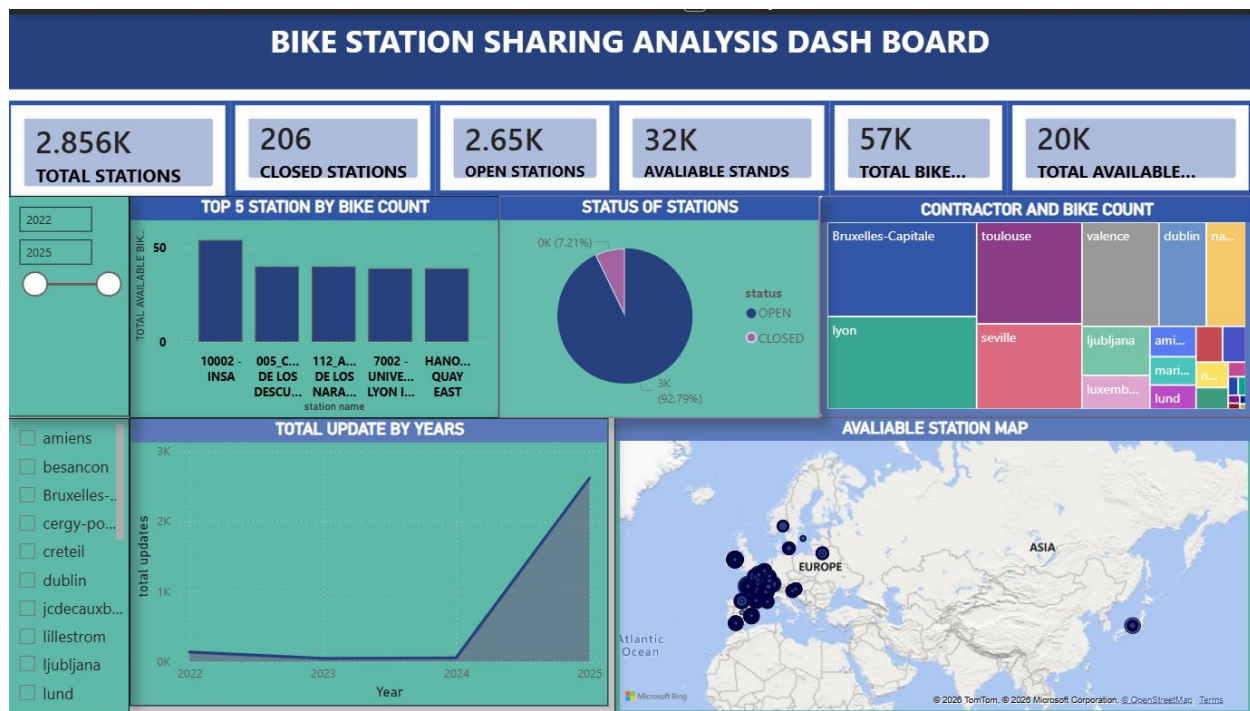
Features Included:

- Filters (Slicers)

➤ Interactive visuals

Multiple Visualizations based on problem statement:

- ❖ Bar chart (top 5 station by bike count)
- ❖ Pie chart (status of stations)
- ❖ Tree map (contractor name and their bike count)
- ❖ Line chart (total updates based on year)
- ❖ Map chart (different availability of bikes across Europe)



8. KEY INSIGHTS (SMART + ANALYTICS APPROACH):

The following KPIs were identified through the analysis.

8. 1. DESCRIPTIVE ANALYTICS:

- Identifies the busiest stations such as **INSA** and **Hanover Quay East**, giving a measurable view of demand concentration and helping prioritize resource allocation.
- Shows that the majority of stations are open while a smaller portion is closed, providing a quick operational health check for service reliability.
- Highlights contractors like **Bruxelles-Capitale**, **Lyon**, and **Dublin** with significant bike counts, making it relevant for evaluating contractor performance and regional coverage.
- Displays steady growth from **2022** to **2025**, with a sharp rise in **2025**, offering a time-bound perspective on expansion and infrastructure updates.
- Visualizes the distribution of stations across **Europe**, ensuring stakeholders can assess accessibility and urban mobility reach in specific regions.

8.2. DIAGNOSTIC ANALYTICS:

- Demand is concentrated in specific hubs such as **INSA** and **Hanover Quay East**. This suggests uneven distribution, where certain stations are over-utilized while others may be underperforming.
- The high proportion of open stations compared to closed ones indicates strong operational reliability. The **206** closed stations may be linked to contractor inefficiencies, maintenance delays, or low-demand areas.
- Contractors like **Bruxelles-Capitale**, **Lyon**, and **Dublin** dominate bike counts, showing strong regional adoption. Smaller contractors may struggle with limited infrastructure or weaker user engagement.
- The sharp rise in **2025** suggests accelerated expansion or upgrades. This could be driven by policy initiatives, increased urban demand, or contractor investments, but may also signal uneven growth across regions.
- The spread across major **European** cities highlights strong coverage, but gaps in certain regions suggest untapped potential or logistical challenges in expanding to less urbanized areas.

8.3. PREDICTIVE ANALYTICS:

- With updates rising sharply in **2025**, the total number of stations is projected to surpass **3K+** by **2026**, indicating continued expansion across major **European** cities.
- Current availability of **20K** bikes suggests demand pressure; predictive trends point to a need for at least **25–30K** available bikes in the next two years to meet rising urban mobility needs.
- Leading contractors such as **Bruxelles-Capitale**, **Lyon**, and **Dublin** are expected to maintain dominance, while smaller contractors may gradually scale up, reducing regional disparities.
- With most stations open, predictive models suggest closures will remain below **10%**, provided maintenance and contractor efficiency are sustained.
- The map indicates strong coverage in **Western Europe**; predictive insights suggest growth opportunities in underrepresented regions like **Eastern Europe** and **Southern Italy**, driven by urbanization and policy support.

8.4. PRESCRIPTIVE ANALYTICS:

- Increase the number of available bikes from **20K** toward **25–30K** to meet rising demand, especially in high-traffic hubs like **INSA** and **Hanover Quay East**.
- Investigate the **206** closed stations and implement proactive maintenance or contractor support programs to keep closures below **10%**.
- Support **smaller contractors** with infrastructure investment and training to reduce disparities, ensuring more uniform coverage across **Europe**.
- Target underrepresented regions (**Eastern Europe, Southern Italy**) for new stations, leveraging urbanization trends and policy incentives to maximize reach.
- Align yearly updates with predictive demand, ensuring sustainable expansion beyond **3K** stations by **2026**, while maintaining operational efficiency and accessibility.

9. CONCLUSION:

The analysis confirms strong growth in bike-sharing across **Europe**, with over **2.8K** stations and reliable operations. High-demand hubs like **INSA** and **Hanover Quay East** highlight the need for increased bike availability, while contractor disparities and closed stations require targeted action. Expansion into underrepresented regions such as **Eastern Europe** and **Southern Italy** will ensure balanced coverage and sustainable urban mobility.